



CS 886 Deep Learning for Biotechnology

Ming Li

CONTENT

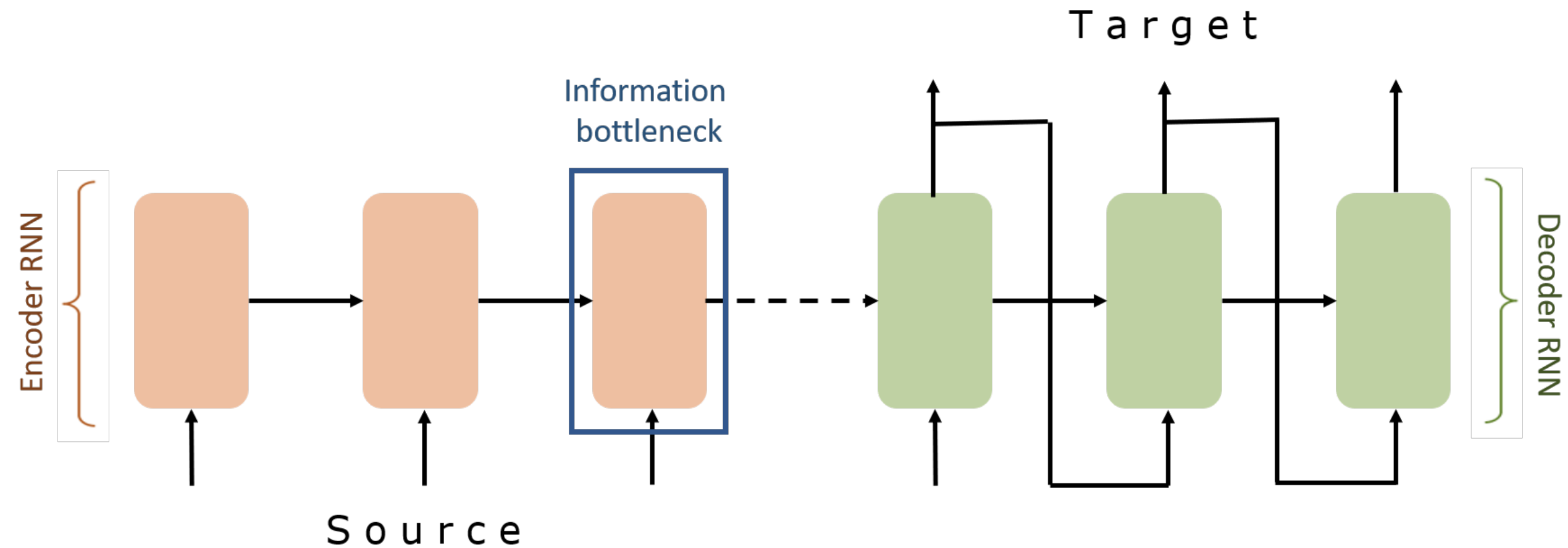
- 01. Word2Vec
- 02. Attention / Transformers
- 03. Pretraining: GPT / BERT
- 04. Deep learning applications in proteomics
- 05. Student presentations begin
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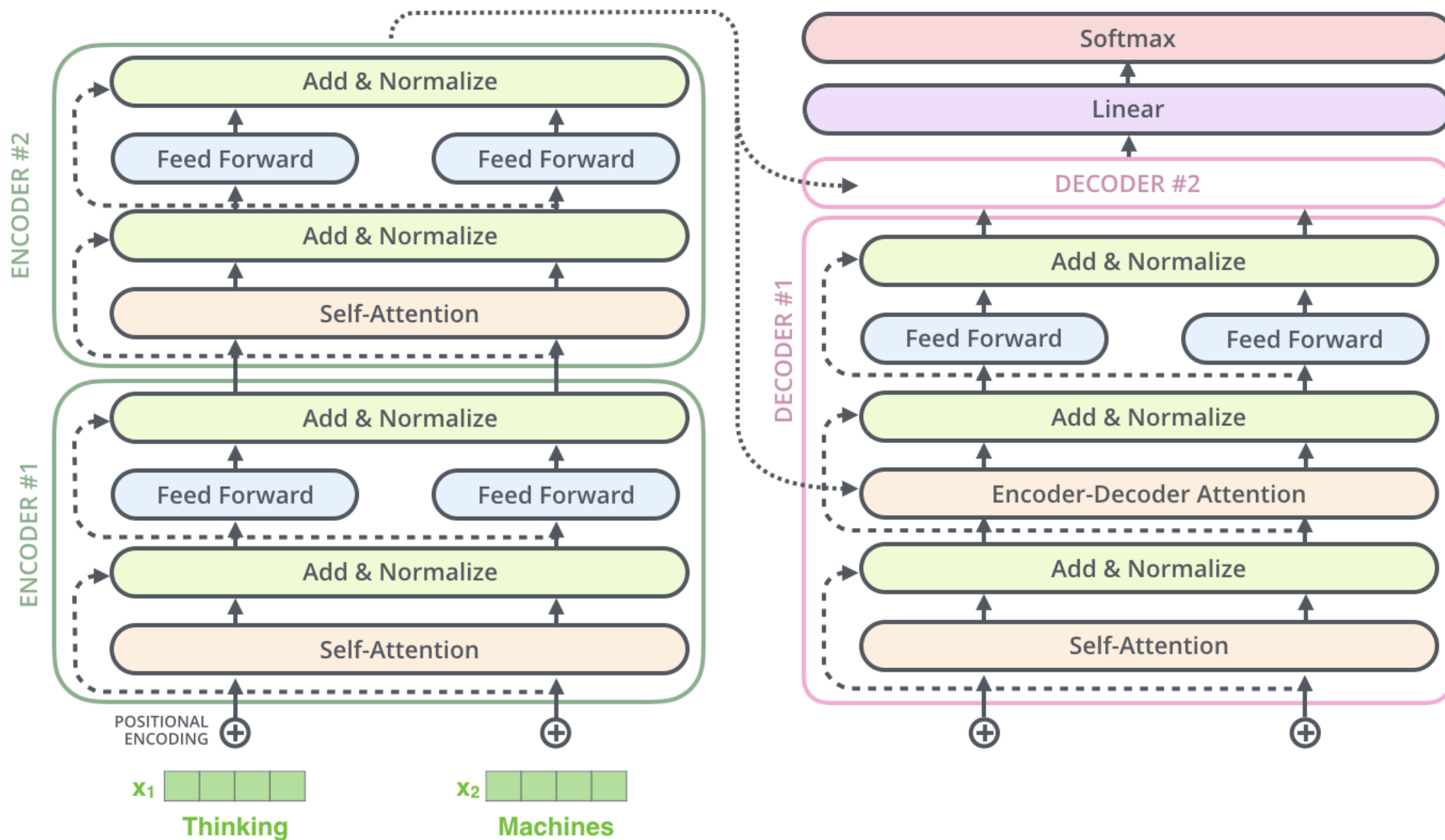


Pretraining: GPT-2 and BERT

LECTURE
THREE

Avoiding Information bottleneck



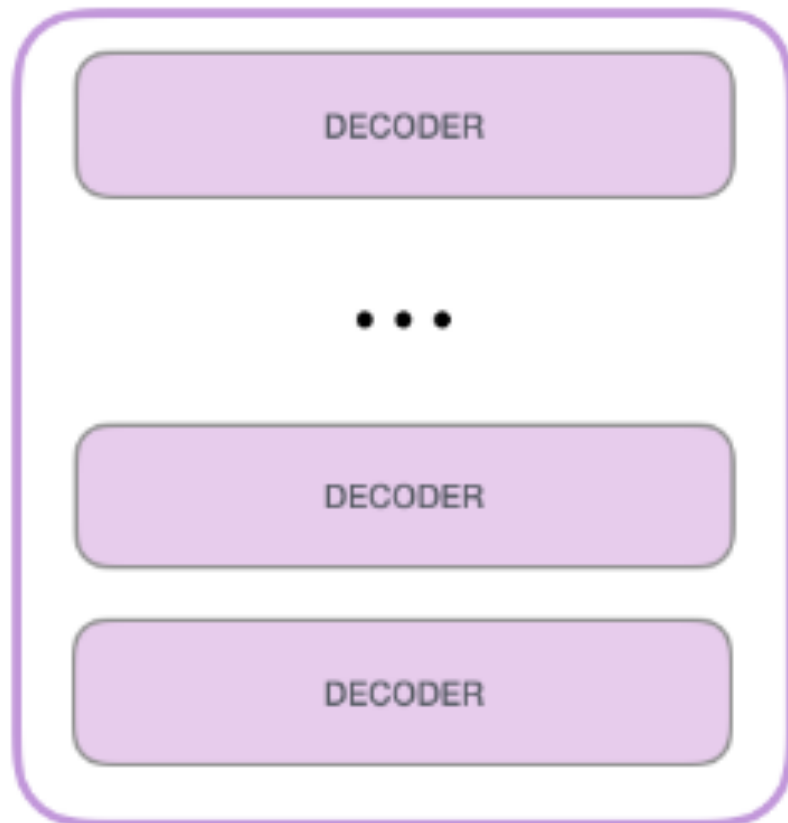


Transformers, GPT-2, and BERT

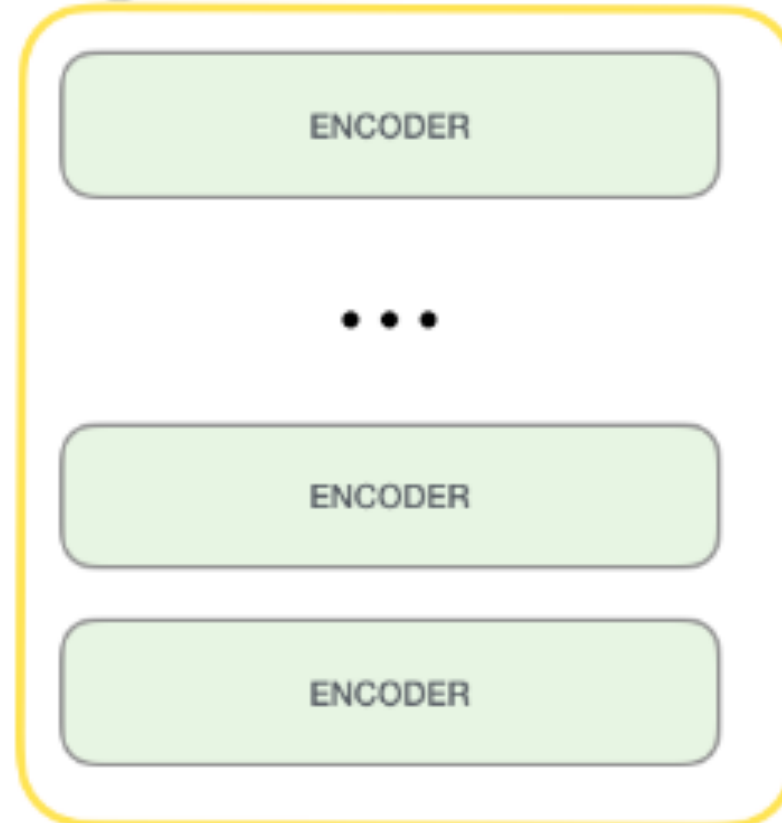
1. A transformer uses Encoder stack to model input, and uses Decoder stack to model output (using input information from encoder side).
2. But if we do not have input, we just want to model the “next word” , we can get rid of the Encoder side of a transformer and output “next word” one by one. This gives us GPT.
3. If we are only interested in training a language model for the input for some other tasks, then we do not need the Decoder of the transformer, that gives us BERT.



GPT-2



BERT



GPT released June 2018

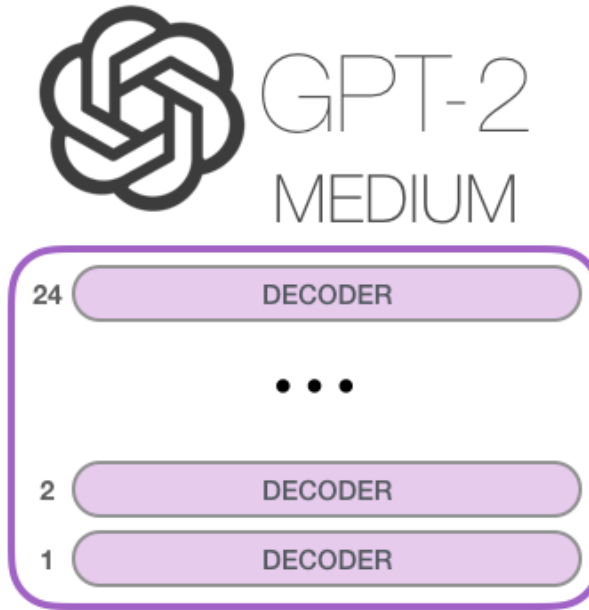
GPT-2 released Nov. 2019 with 1.5B parameters

GPT-3: 175B parameters trained on 45TB texts



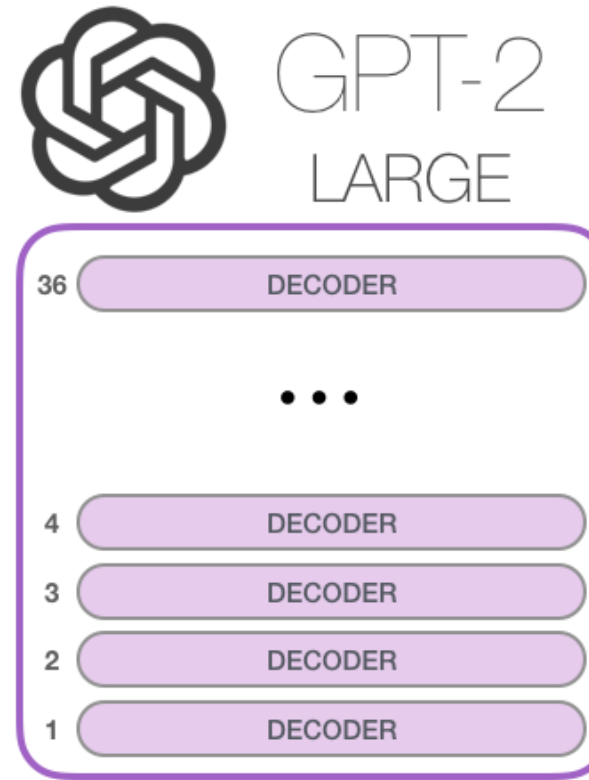
Model Dimensionality: 768

117M parameters



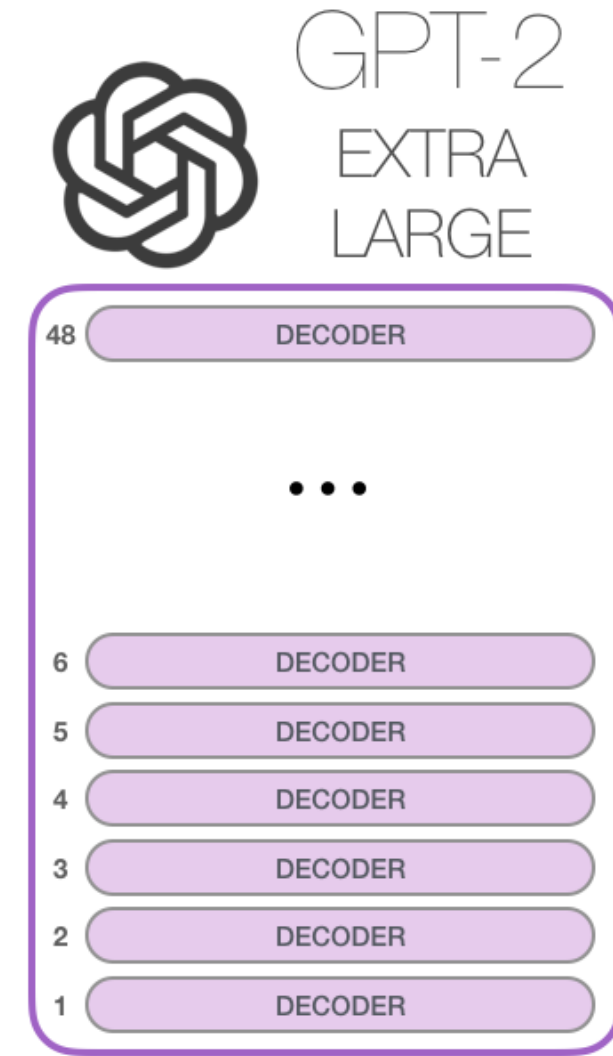
Model Dimensionality: 1024

345M



Model Dimensionality: 1280

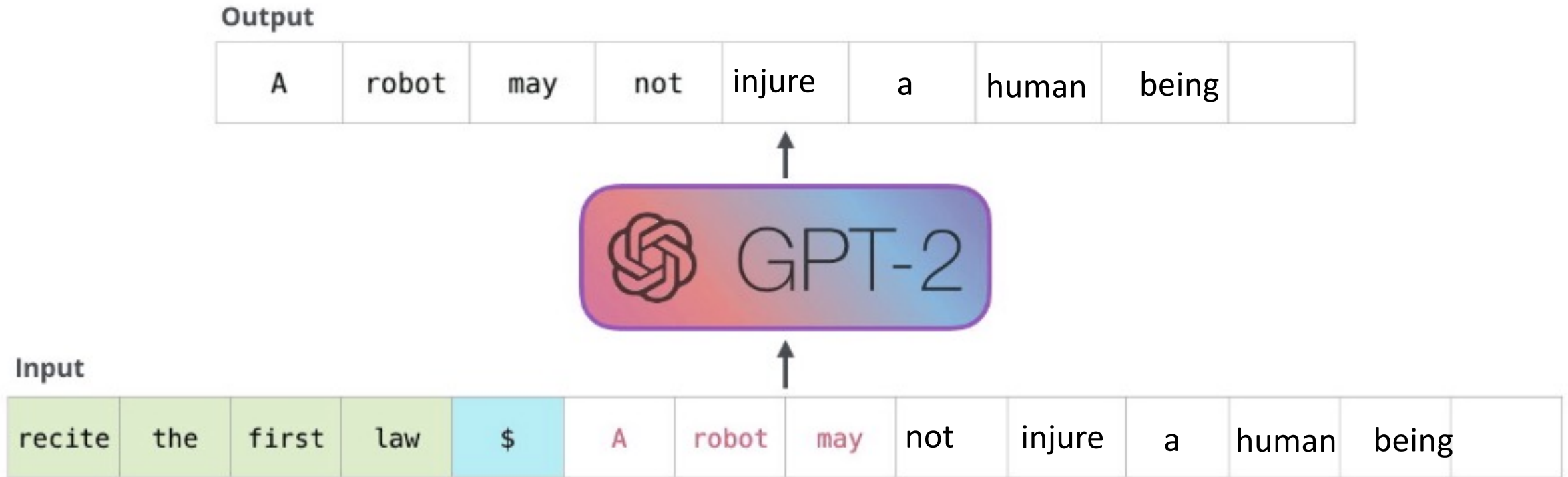
762M



Model Dimensionality: 1600

1542M

GPT-2 in action





Byte Pair Encoding (BPE)

Word embedding sometimes is too high level, pure character embedding too low level. For example, if we have learned

old older oldest

We might also wish the computer to infer

smart smarter smartest

But at the whole word level, this might not be so direct. Thus the idea is to break the words up into pieces like er, est, and embed frequent fragments of words.

GPT adapts this BPE scheme.



Byte Pair Encoding (BPE)

GPT uses BPE scheme. The subwords are calculated by:

1. Split word to sequence of characters (add `</w>` char)
2. Joining the highest frequency pattern.
3. Keep doing step 2, until it hits the pre-defined maximum number of sub-words or iterations.

Example (5, 2, 6, 3 are number of occurrences)

{`'l o w </w>': 5, 'l o w e r </w>': 2, 'n e w e s t </w>': 6, 'w i d e s t </w>': 3 }`

{`'l o w </w>': 5, 'l o w e r </w>': 2, 'n e w e s t </w>': 6, 'w i d e s t </w>': 3 }`

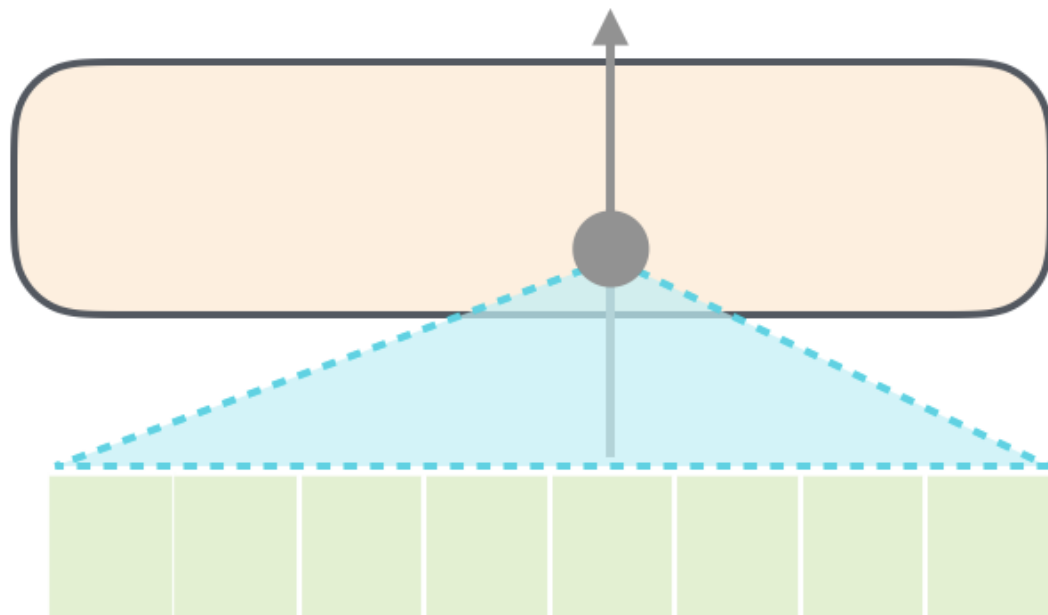
{`'l o w </w>': 5, 'l o w e r </w>': 2, 'n e w e s t </w>': 6, 'w i d e s t </w>': 3 }` (est freq. 9)

{`'lo w </w>': 5, 'lo w e r </w>': 2, 'n e w e s t </w>': 6, 'w i d e s t </w>': 3 }` (lo freq 7)

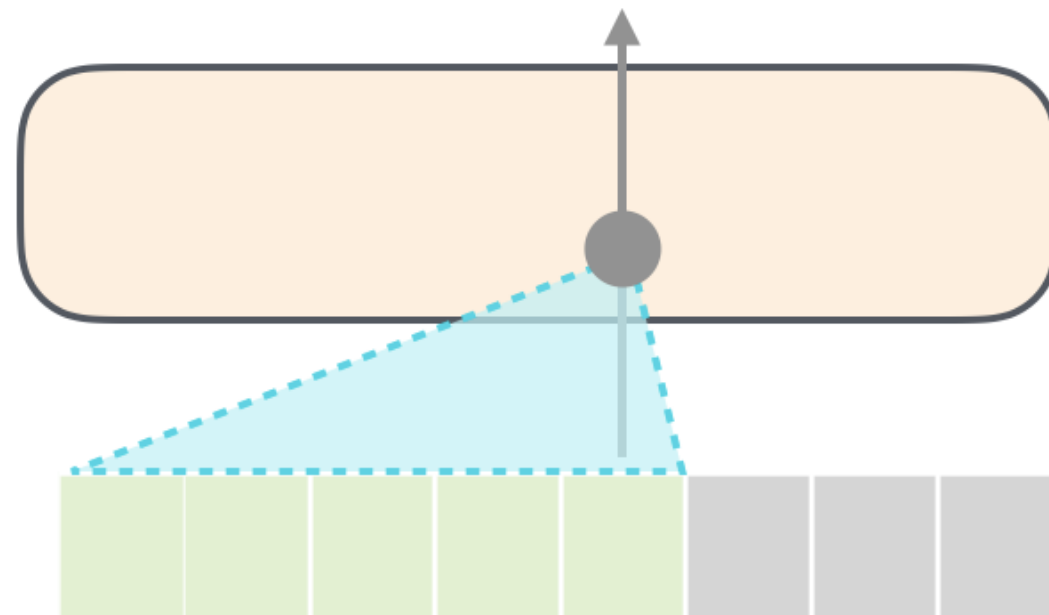
.....

Masked Self-Attention (to compute more efficiently)

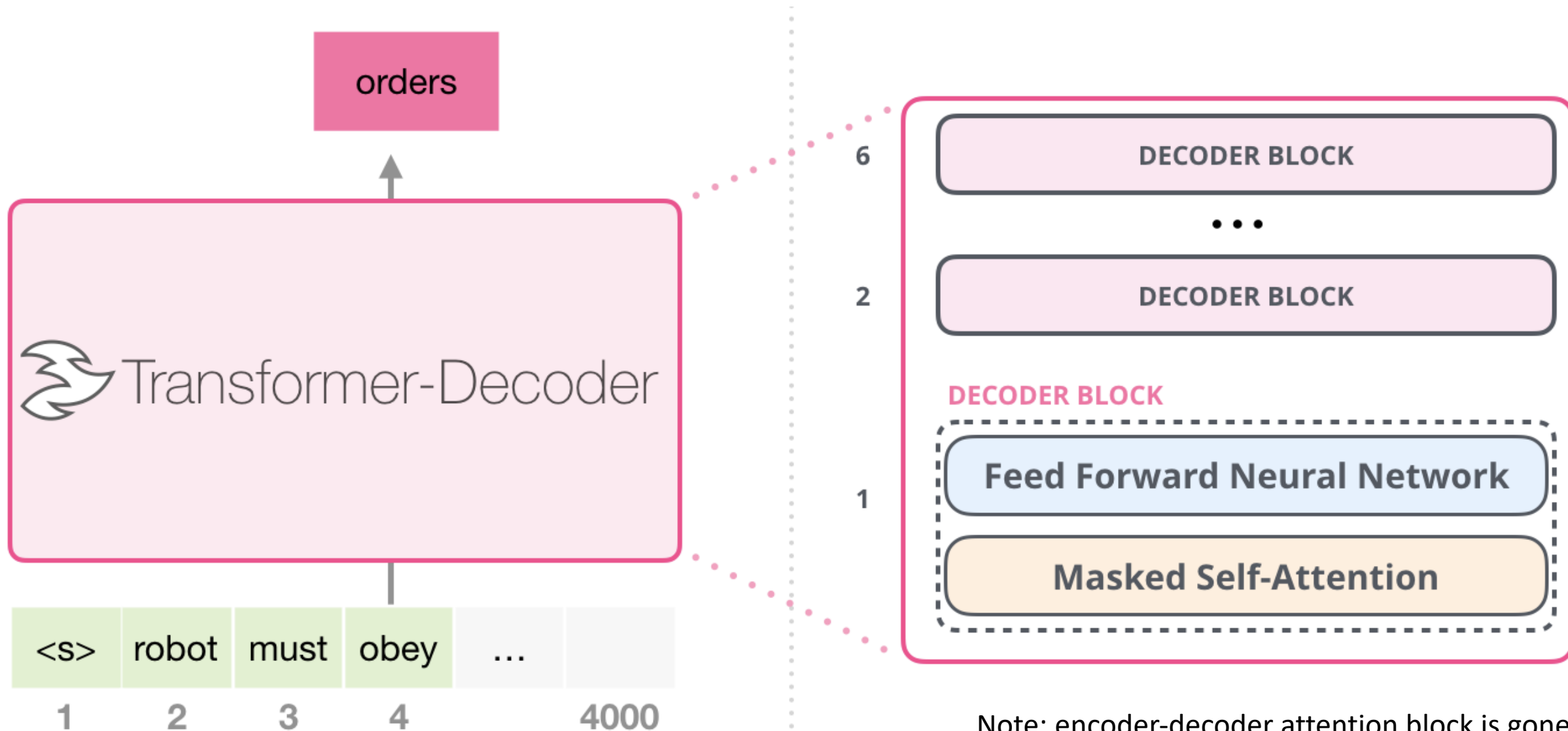
Self-Attention



Masked Self-Attention



Masked Self-Attention

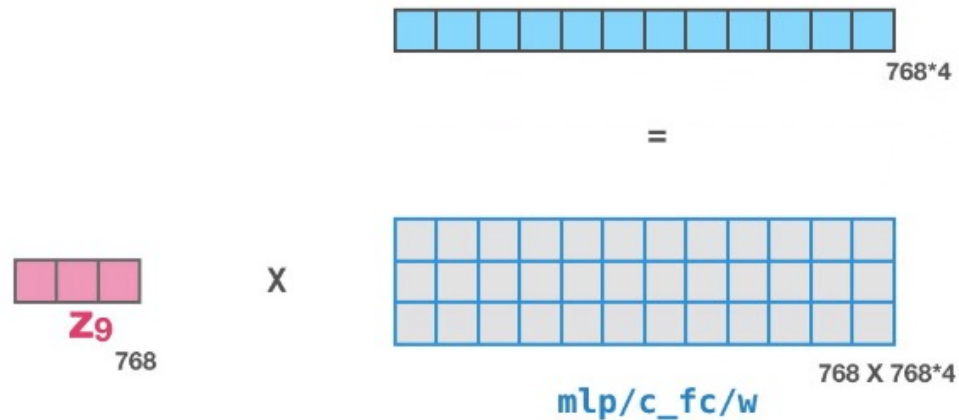


Masked Self-Attention Calculation

Re-use previous computation results: at any step, only need to results of q , k , v related to the new output word, no need to re-compute the others. Additional computation is linear, instead of quadratic.

GPT-2 fully connected network has two layers (Example for GPT-2)

GPT2 Fully-Connected Neural Network

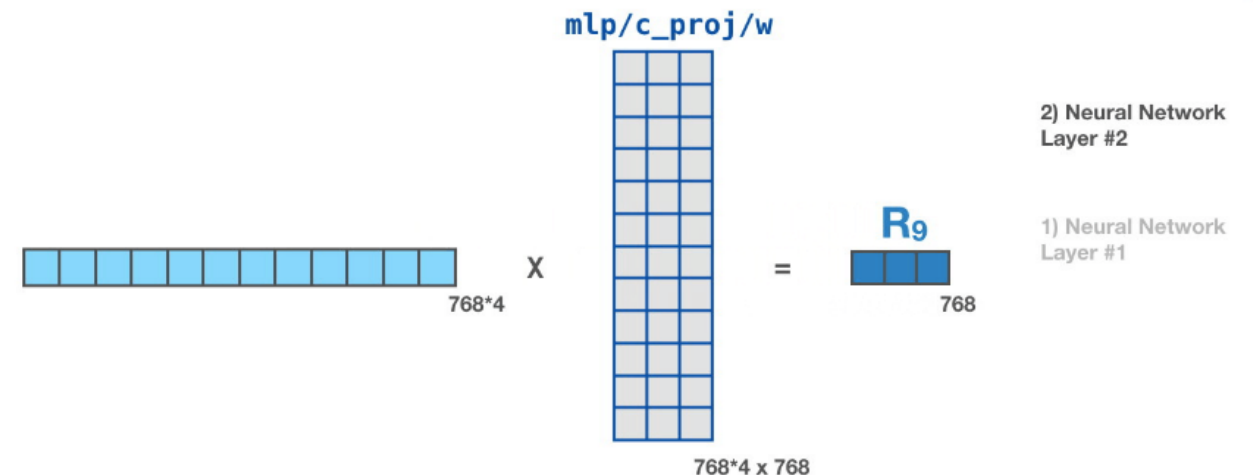


2)

1) Neural Network Layer #1

768 is small model size

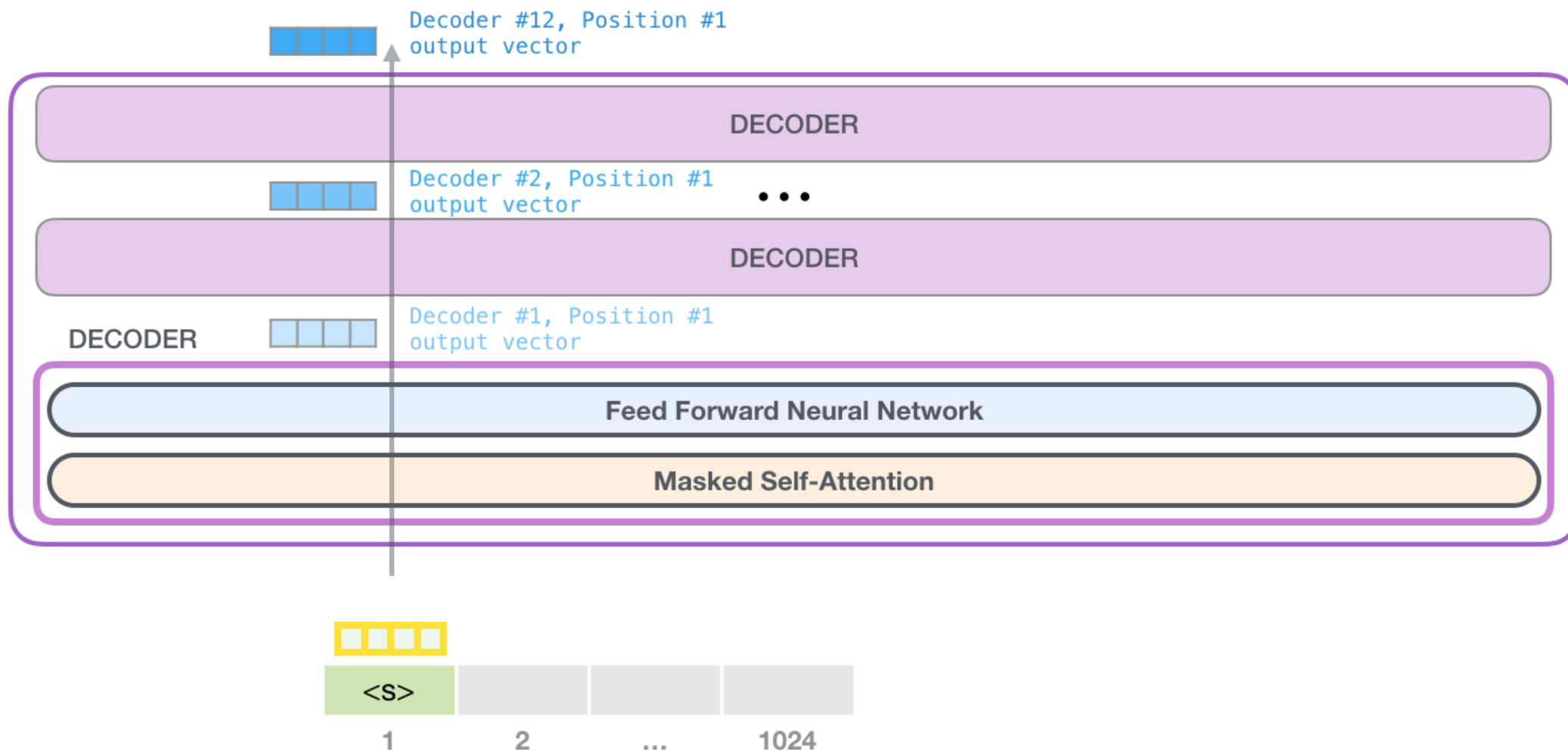
GPT2 Fully-Connected Neural Network



2) Neural Network Layer #2

1) Neural Network Layer #1

GPT-2, BERT
— GPT-2 has a parameter top-k, so that we sample words from top k (highest probability from softmax) words for each output



[illegible]

GPT Training

GPT-2 uses **unsupervised** learning approach to training the language model.

There is no custom training for GPT-2, no separation of pre-training and fine-tuning like BERT.

A story generated by GPT-2

“The scientist named the population, after their distinctive horn, Ovid’s Unicorn. These four-horned, silver-white unicorns were previously unknown to science.

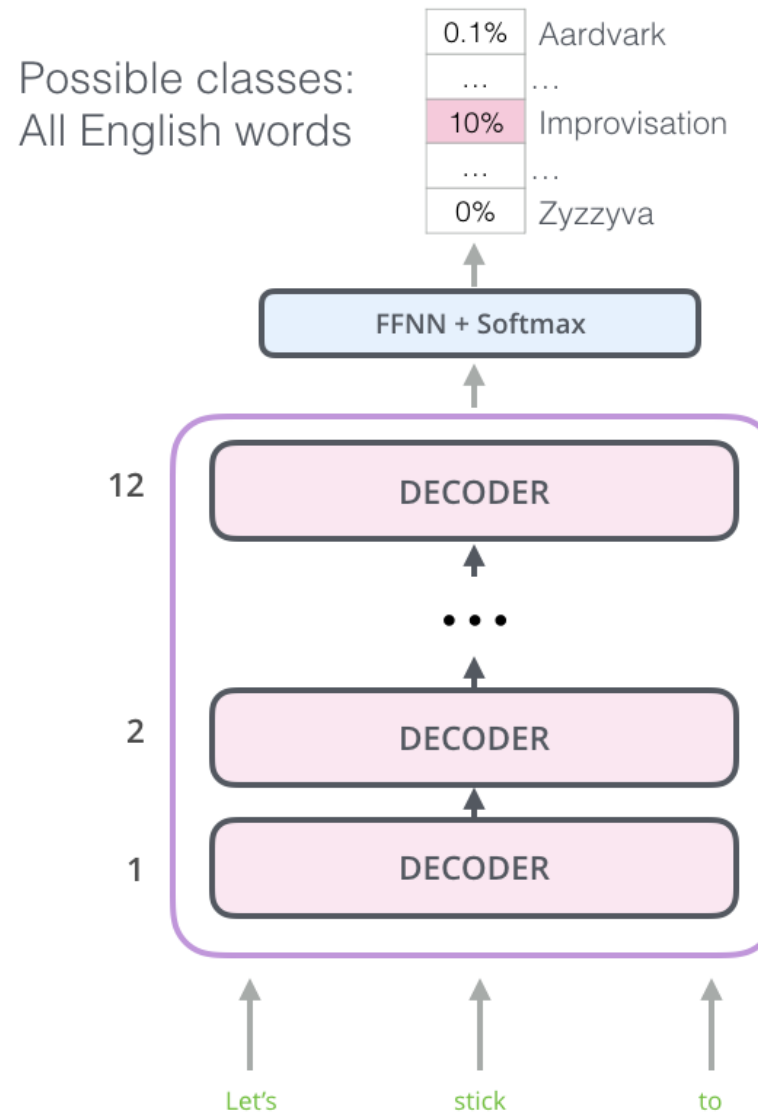
Now, after almost two centuries, the mystery of what sparked this odd phenomenon is finally solved.

Dr. Jorge Pérez, an evolutionary biologist from the University of La Paz, and several companions, were exploring the Andes Mountains when they found a small valley, with no other animals or humans. Pérez noticed that the valley had what appeared to be a natural fountain, surrounded by two peaks of rock and silver snow.

Pérez and the others then ventured further into the valley. ‘By the time we reached the top of one peak, the water looked blue, with some crystals on top,’ said Pérez.

Pérez and his friends were astonished to see the unicorn herd. These creatures could be seen from the air without having to move too much to see them – they were so close they could touch their horns."

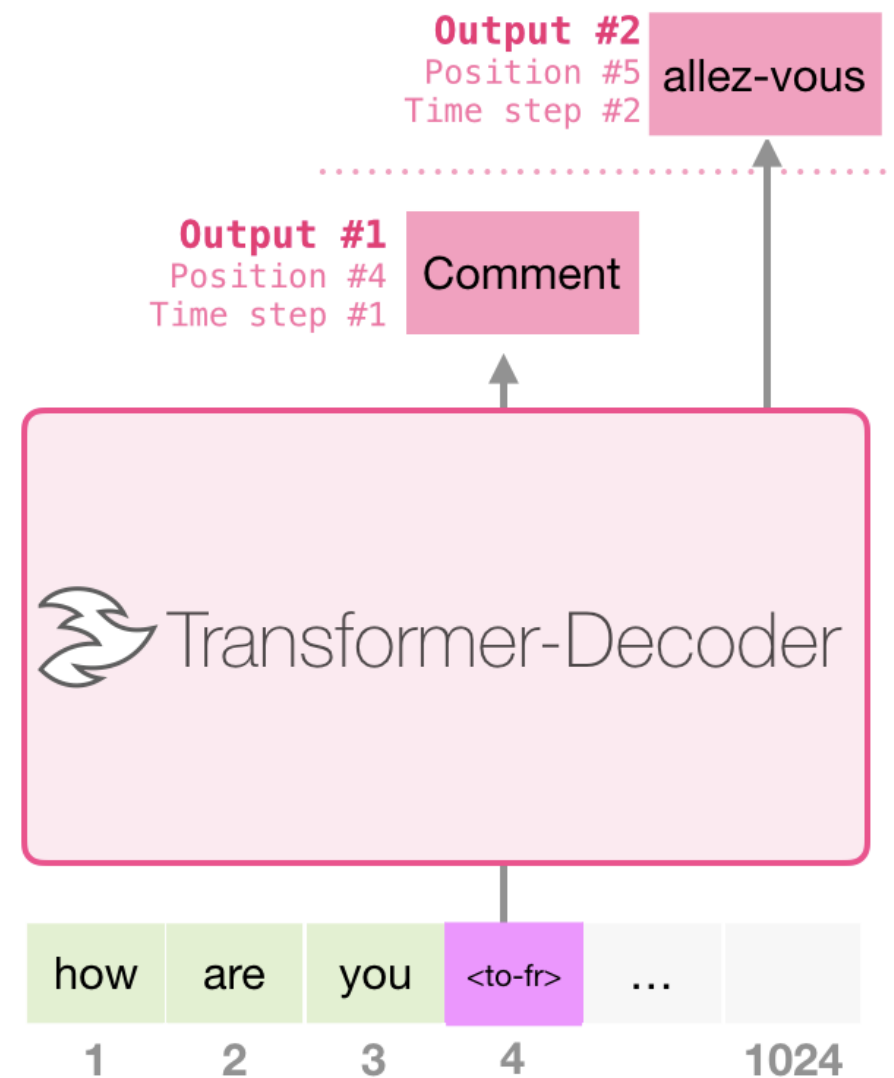
Transformer / GPT prediction



GPT-2 Application: Translation

Training Dataset

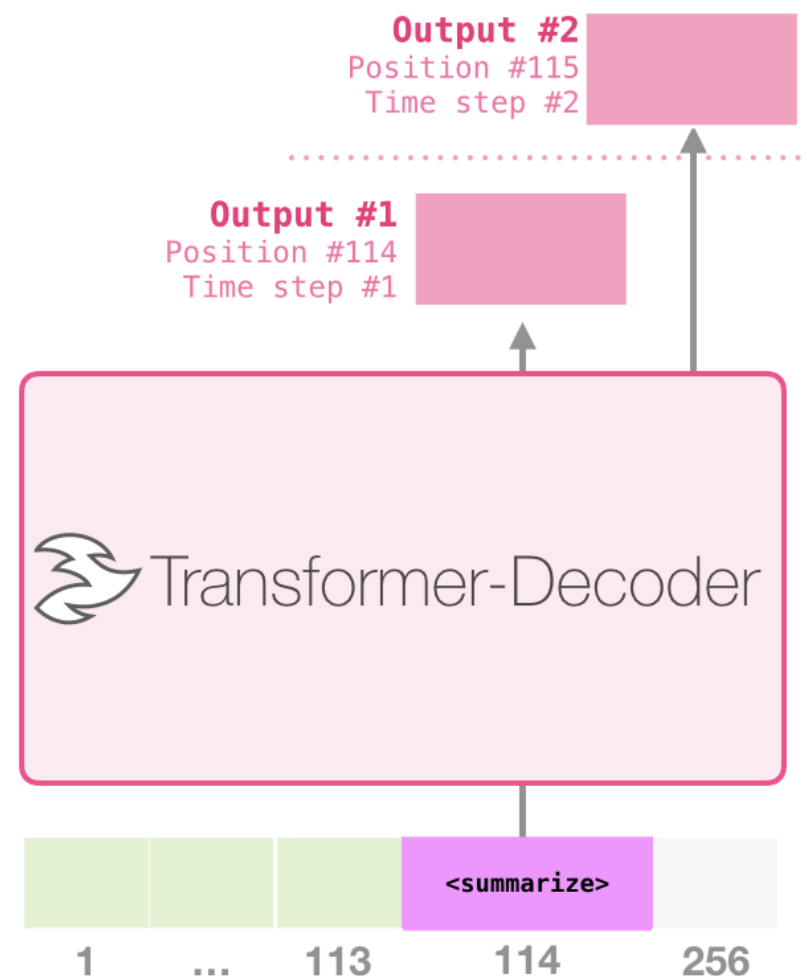
I	am	a	student	<to-fr>	je	suis	étudiant
let	them	eat	cake	<to-fr>	Qu'ils	mangent	de
good	morning	<to-fr>	Bonjour				



GPT-2 Application: Summarization

Training Dataset

Article #1 tokens	<summarize>	Article #1 Summary	
Article #2 tokens	<summarize>	Article #2 Summary	padding
Article #3 tokens	<summarize>	Article #3 Summary	





Using wikipedia data

WIKIPEDIA

The Free Encyclopedia

Main page

Contents

Featured content

Current events

Random article

Donate to Wikipedia

Wikipedia store

Interaction

Help

About Wikipedia

Community portal

Recent changes

Contact page

Tools

What links here

Related changes

Upload file

Special pages

Permanent link

Page information

Wikidata item

Cite this page

Print/export

Create a book

Download as PDF

Portable version

Languages

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Spanish

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Italian

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Portuguese

Swedish

Turkish

33,300 more

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Talk

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Create account

Log in

Read

Edit

View history

Search Wikipedia

Positronic brain

From Wikipedia, the free encyclopedia
(redirected from Positronic robot)

This article is about a fictional technological device. For the manufacturing company based in Springfield, Missouri, see *Positronic company*.

This article needs additional citations for verification. Please help improve this article by adding citations to reliable sources. Unsourced material may be challenged and removed.

Find sources: "Positronic brain" – news – newspapers – books – scholar – JSTOR (July 2008) (Learn how and when to remove this template message)

A **positronic brain** is a fictional technological device, originally conceived by science fiction writer Isaac Asimov^[1]. It functions as a central processing unit (CPU) for robots, and, in some unspecified way, provides them with a form of consciousness recognizable to humans. When Asimov wrote his first robot stories in 1939 and 1940, the positron was a newly discovered particle, and so the buzz word *positronic* added a contemporary gloss of popular science to the concept. The short story "Runaround", by Asimov, elaborates on the concept, in the context of his fictional Three Laws of Robotics.

Contents

1 Conceptual overview

2 In Allen's trilogy

3 References in other fiction and films

3.1 Abbott and Costello Go To Mars

3.2 The Avengers

3.3 Doctor Who

3.4 Star Trek

3.5 Perry Rhodan

3.6 I, Robot, 2004 Film

3.7 Bicentennial Man

3.8 Buck Rogers in the 25th Century

3.9 Mystery Science Theater 3000

3.10 Spectreman

3.11 Stellan

4 References

5 External links

Conceptual overview

Asimov remained vague about the technical details of positronic brains except to assert that their substructure was formed from an alloy of *platinum and indium*. They were said to be vulnerable to radiation and apparently involve a type of *volatile memory* (since robots in storage required a power source keeping their brains "alive"). The focus of Asimov's stories was directed more towards the *software* of robots—such as the Three Laws of Robotics—than the hardware in which it was implemented, although it is stated in his stories that to create a positronic brain without the Three Laws, it would have been necessary to spend years redesigning the fundamental approach towards the brain itself.

Within his stories of *robotics* on Earth and their development by U.S. Robots, Asimov's positronic brain is less of a *plot device* and more of a technological item worthy of study.

A positronic brain cannot ordinarily be built without incorporating the Three Laws; any modification thereof would drastically modify robot behavior. Behavioral dilemmas resulting from conflicting potentials set by inexperienced and/or malicious users of the robot for the Three Laws make up the bulk of Asimov's stories concerning robots. They are resolved by applying the *science* of logic and *psychology* together with *mathematics*, the supreme solution finder being Dr. Susan Calvin, Chief Roboticspsychologist of U.S. Robots.

The Three Laws are also a *cornerstone* in brain sophistication. Very complex brains designed to handle world economy interpret the First Law in expanded sense to include humanity as opposed to a single human; in Asimov's later works *the Robots and Empire* this is referred to as the "Zenith Law". At least one brain constructed as a calculating machine, as opposed to being a robot control circuit, was designed to have a flexible, childlike personality so that it was able to pursue difficult problems without the Three Laws inhibiting it completely. Specialized brains created for overseeing world economics were stated to have no personality at all.

Under specific conditions, the Three Laws can be obviated, with the modification of the actual robotic design.

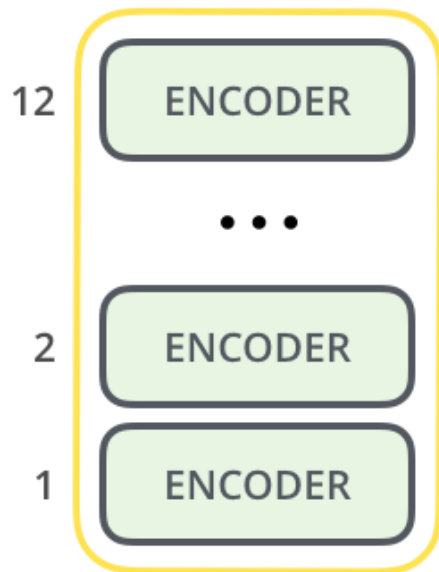
- Robots that are of low enough value can have the **Third Law** deleted; they do not have to protect themselves from harm, and the brain size can be reduced by half.
- Robots that do not require orders from a human being may have the **Second Law** deleted, and therefore require smaller brains again, providing they do not require the Third Law.
- Robots that are disposable, cannot receive orders from a human being and are not able to harm a human, will not require even the **First Law**. The sophistication of positronic circuitry renders a brain so small that it could comfortably fit within the skull of an insect.

Robots of the latter type directly parallel contemporary industrial robotics practice, though real-life robots do contain safety sensors and systems, in a concern for human safety (a weak form of the First Law; the robot is a safe tool to use, but has no "judgment", which is implicit in Asimov's own stories).

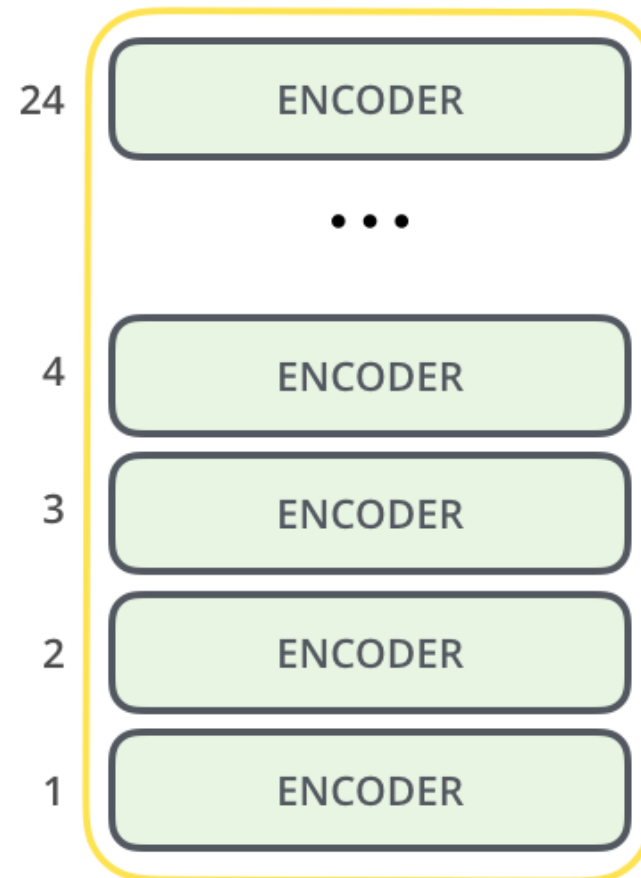
In Allen's trilogy

Several robot stories have been written by other authors following Asimov's death. For example, in Roger MacBride Allen's *Caliban* trilogy, a Spacer robotist called Gubler Anshaw invents the **gravitonic brain**. It offers speed and capacity improvements over traditional positronic designs, but the strong influence of tradition make robotics labs reject Anshaw's work. Only one robotist, Freda Leving, chooses to adopt gravitronics, because it offers her a blank slate on which she could explore alternatives to the Three Laws. Because they are not dependent upon centuries of earlier research, gravitonic brains can be programmed with the standard Laws, variations of the Laws, or even empty pathways which specify no Laws at all.

BERT (Bidirectional Encoder Representation from Transformers)



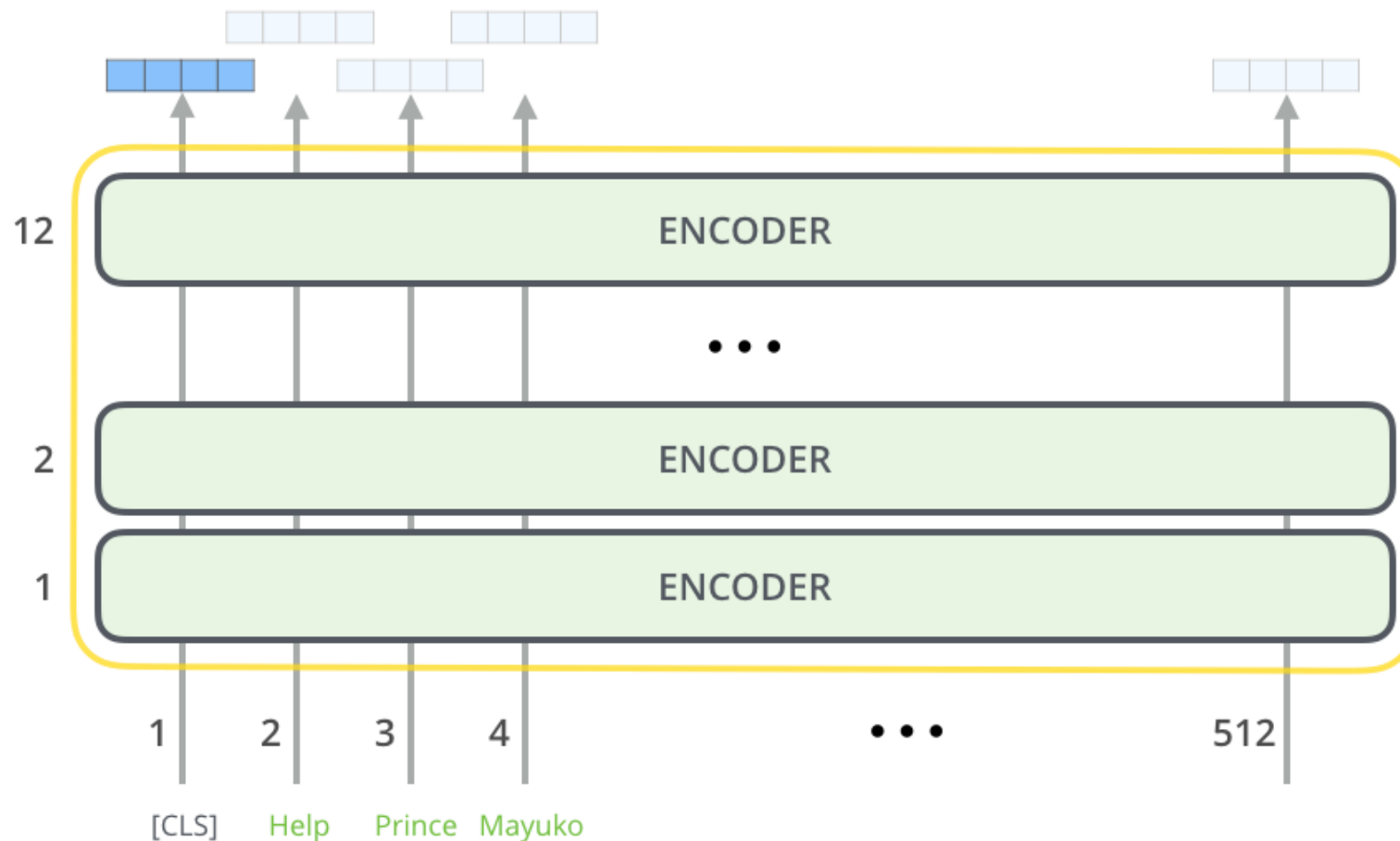
BERT_{BASE}



BERT_{LARGE}

Model input dimension 512

Input and output vector size



BERT

BERT pretraining

ULM-FiT (2018): Pre-training ideas, transfer learning in NLP.

ELMo: Bidirectional training (LSTM)

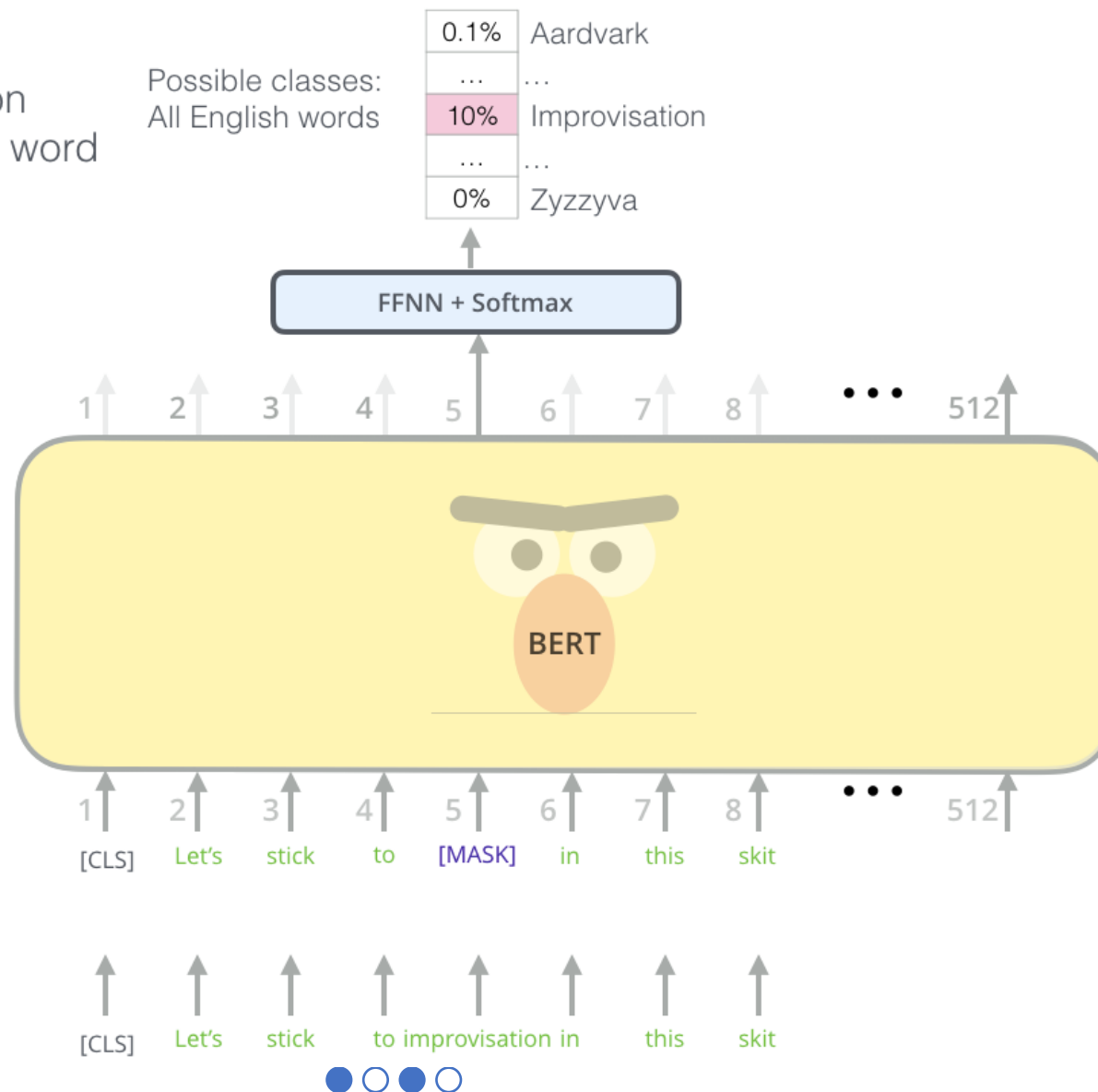
Transformer: Although used things from left, but still missing from the right.

GPT: Use Transformer Decoder half.

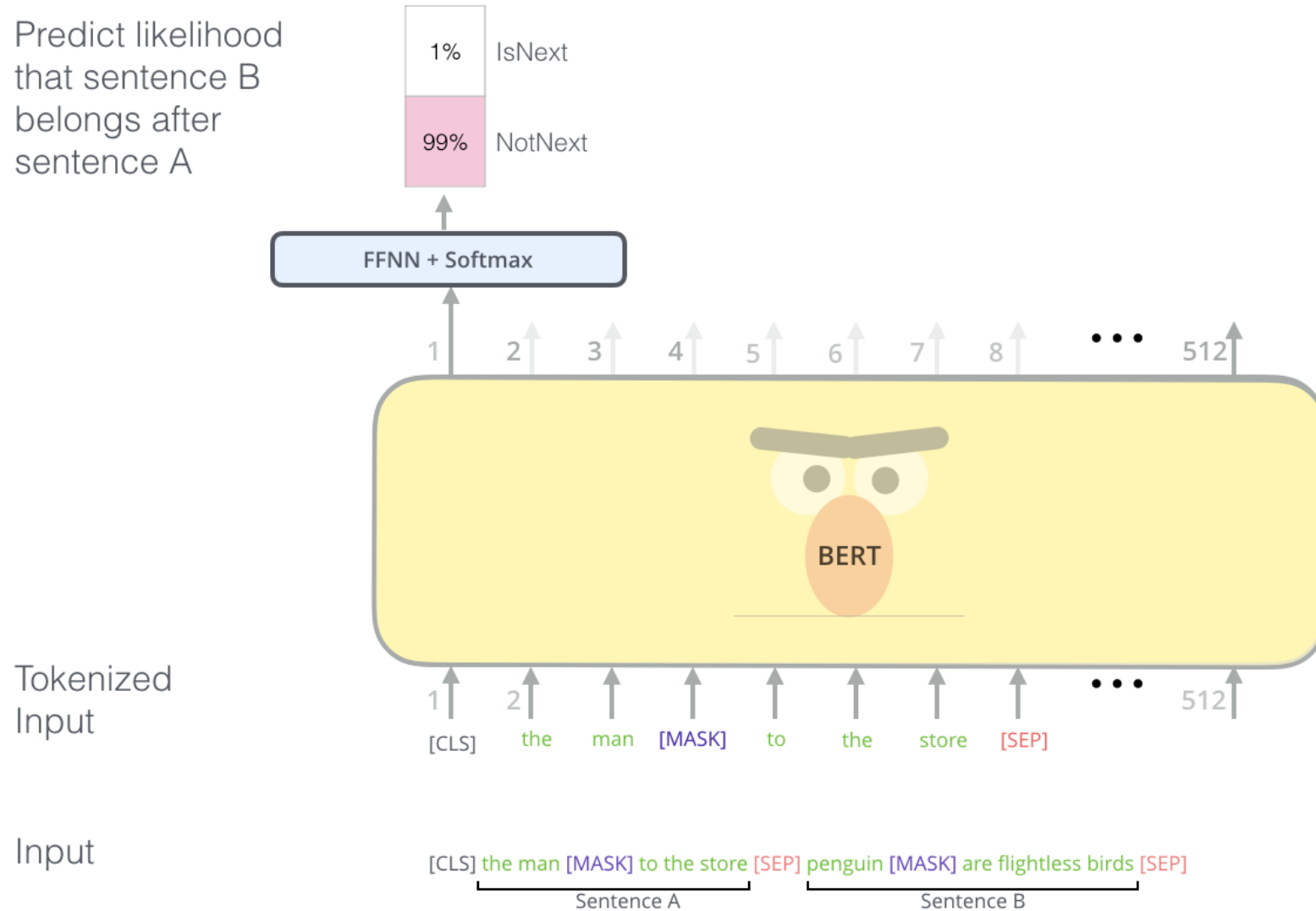
BERT: Switches from Decoder to Encoder, so that it can use both sides in training and invented corresponding training tasks: masked language model

BERT Pretraining Task 1: masked words

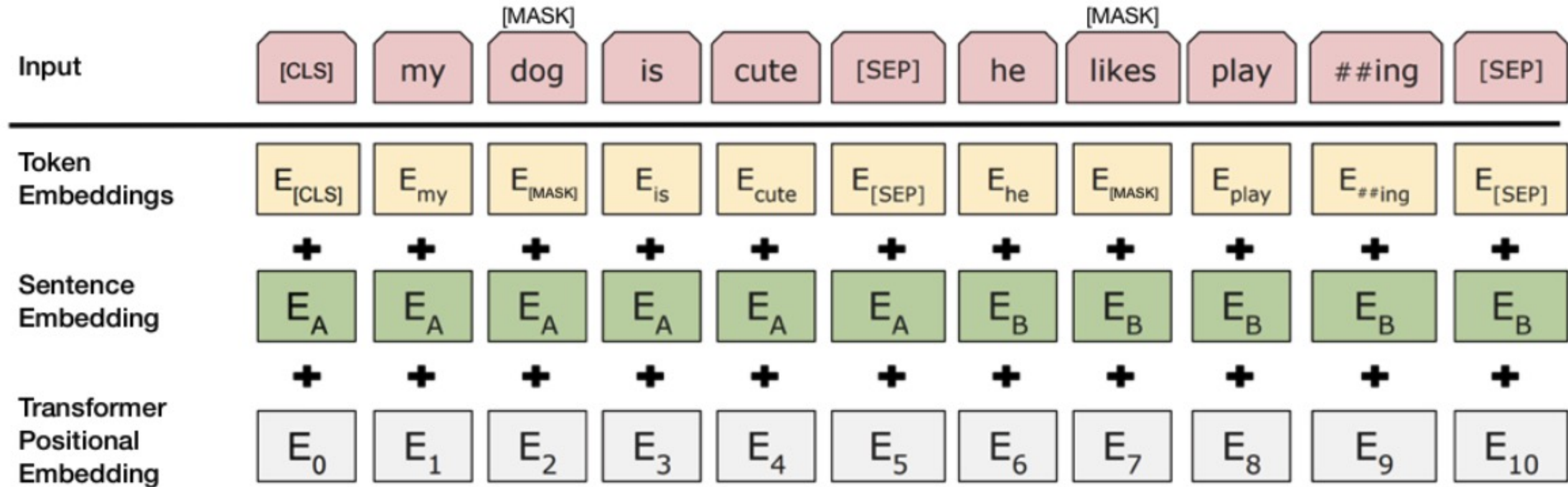
Use the output of the masked word's position to predict the masked word



BERT Pretraining Task 2: two sentences



BERT Pretraining Task 2: two sentences

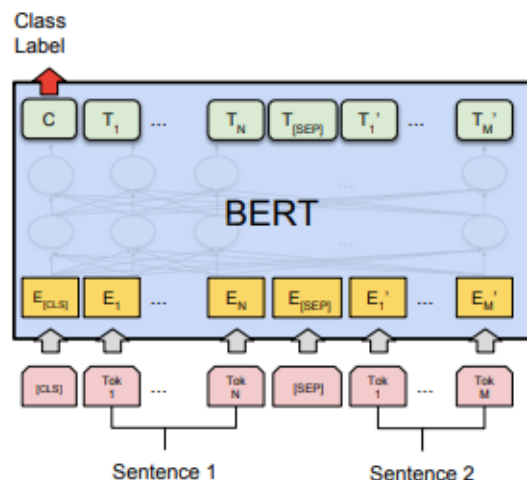


50% true second sentences

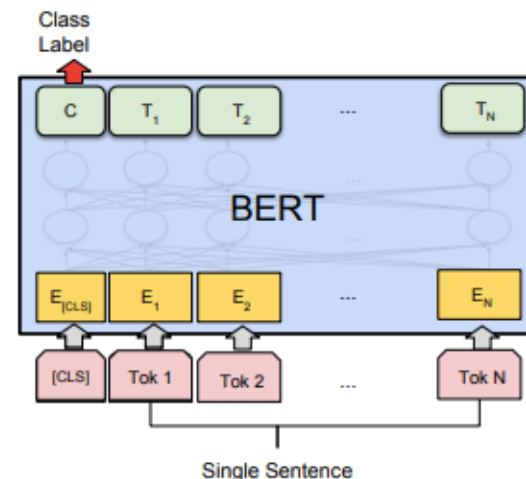
50% random second sentences

Fine-tuning BERT for other specific tasks

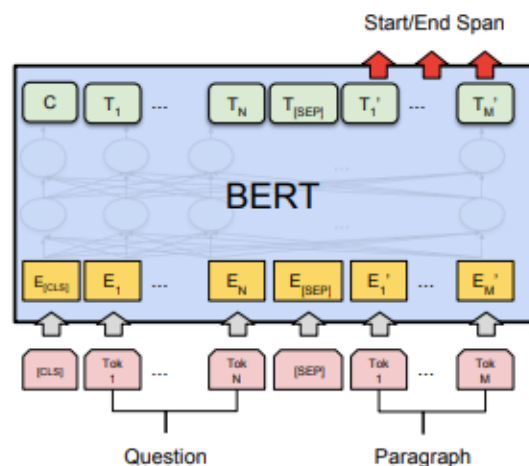
MNLI
 QQP (Quora Question Pairs)
 Semantic equivalence)
 QNLI (NL inference dataset)
 STS-B (text similarity)
 MRPC (paraphrase, Microsoft)
 RTE (textual entailment)
 SWAG (commonsense inference)
 SST-2 (sentiment)
 CoLA (linguistic acceptability)
 SQuAD (question and answer)



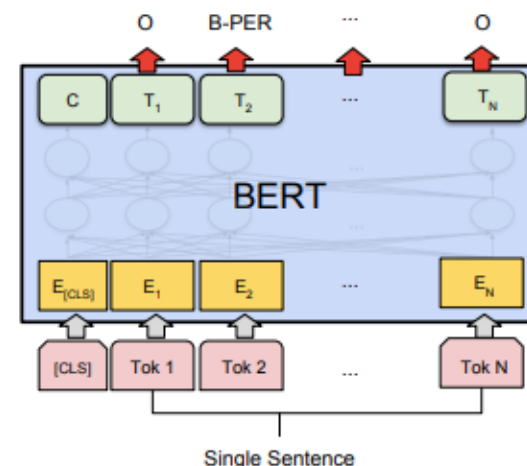
(a) Sentence Pair Classification Tasks:
MNLI, QQP, QNLI, STS-B, MRPC,
RTE, SWAG



(b) Single Sentence Classification Tasks:
SST-2, CoLA



(c) Question Answering Tasks:
SQuAD v1.1



(d) Single Sentence Tagging Tasks:
CoNLL-2003 NER

SST (Stanford
 sentiment treebank):
 215k phrases with
 fine-grained
 sentiment labels in
 the parse trees of
 11k sentences.

NLP Tasks: Multi-Genre Natural Lang. Inference

MNLI: 433k
pairs of
examples,
labeled by
entailment,
neutral or
contraction

Met my first girlfriend that way.	FACE-TO-FACE contradiction C C N C	I didn't meet my first girlfriend until later.
8 million in relief in the form of emergency housing.	GOVERNMENT neutral N N N N	The 8 million dollars for emergency housing was still not enough to solve the problem.
Now, as children tend their gardens, they have a new appreciation of their relationship to the land, their cultural heritage, and their community.	LETTERS neutral N N N N	All of the children love working in their gardens.
At 8:34, the Boston Center controller received a third transmission from American 11	9/11 entailment E E E E	The Boston Center controller got a third transmission from American 11.
I am a lacto-vegetarian.	SLATE neutral N N E N	I enjoy eating cheese too much to abstain from dairy.
someone else noticed it and i said well i guess that's true and it was somewhat melodious in other words it wasn't just you know it was really funny	TELEPHONE contradiction C C C C	No one noticed and it wasn't funny at all.

Table 1: Randomly chosen examples from the development set of our new corpus, shown with their genre labels, their selected gold labels, and the validation labels (abbreviated E, N, C) assigned by individual annotators.



NLP Tasks (SQuAD -- Stanford Question Answering Dataset):

Sample: Super Bowl 50 was an American football game to determine the champion of the National Football League (NFL) for the 2015 season. The American Football Conference (AFC) champion Denver Broncos defeated the National Football Conference (NFC) champion Carolina Panthers 24–10 to earn their third Super Bowl title. The game was played on February 7, 2016, at Levi's Stadium in the San Francisco Bay Area at Santa Clara, California. As this was the 50th Super Bowl, the league emphasized the "golden anniversary" with various gold-themed initiatives, as well as temporarily suspending the tradition of naming each Super Bowl game with Roman numerals (under which the game would have been known as "Super Bowl L"), so that the logo could prominently feature the Arabic numerals 50.

Which NFL team represented the AFC at Super Bowl 50?

Ground Truth Answers: Denver Broncos

Which NFL team represented the NFC at Super Bowl 50?

Ground Truth Answers: Carolina Panthers

Add indices for sentences and paragraphs

SegaTron/SegaBERT

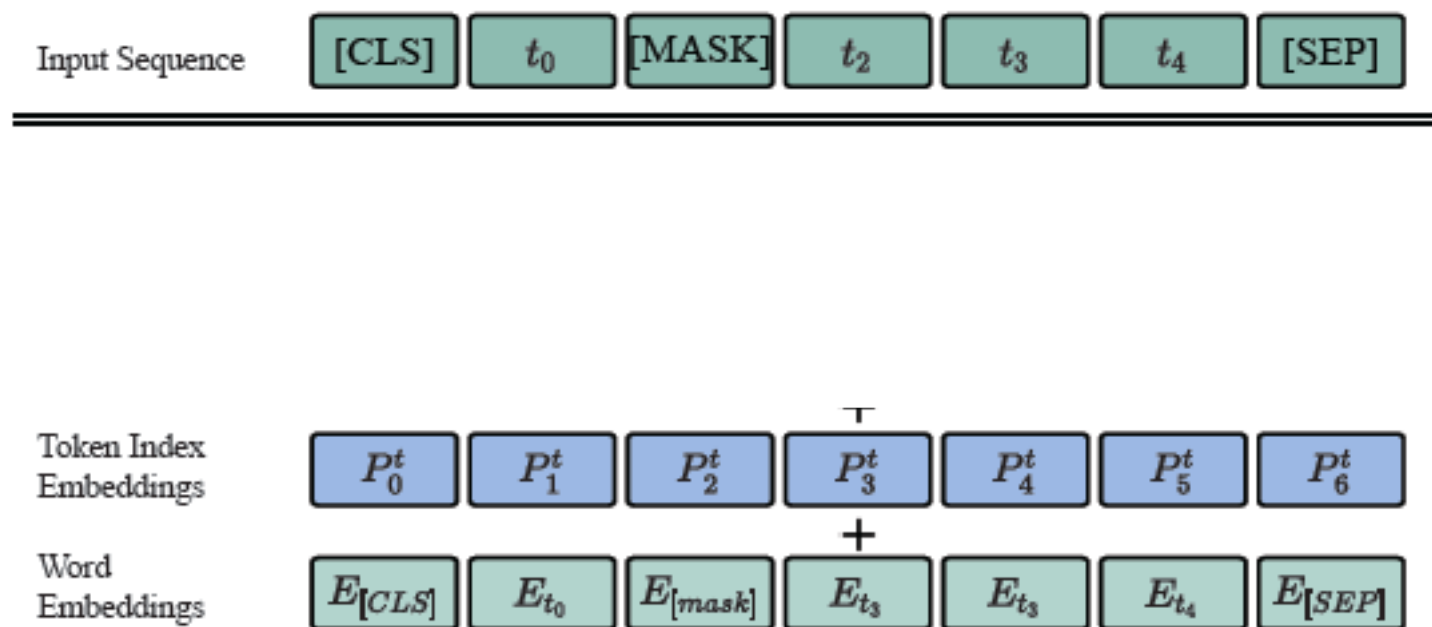
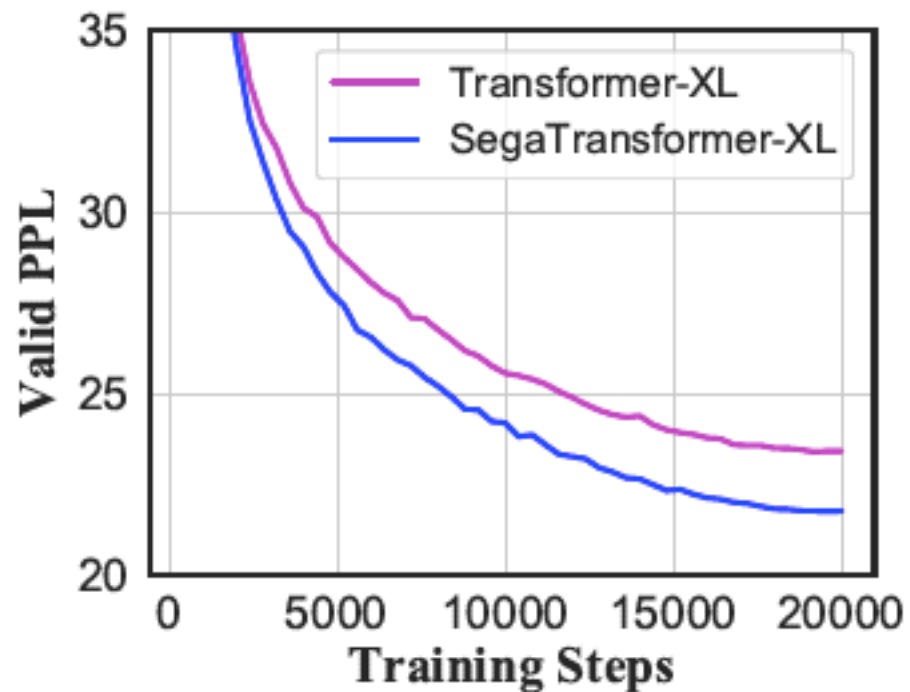
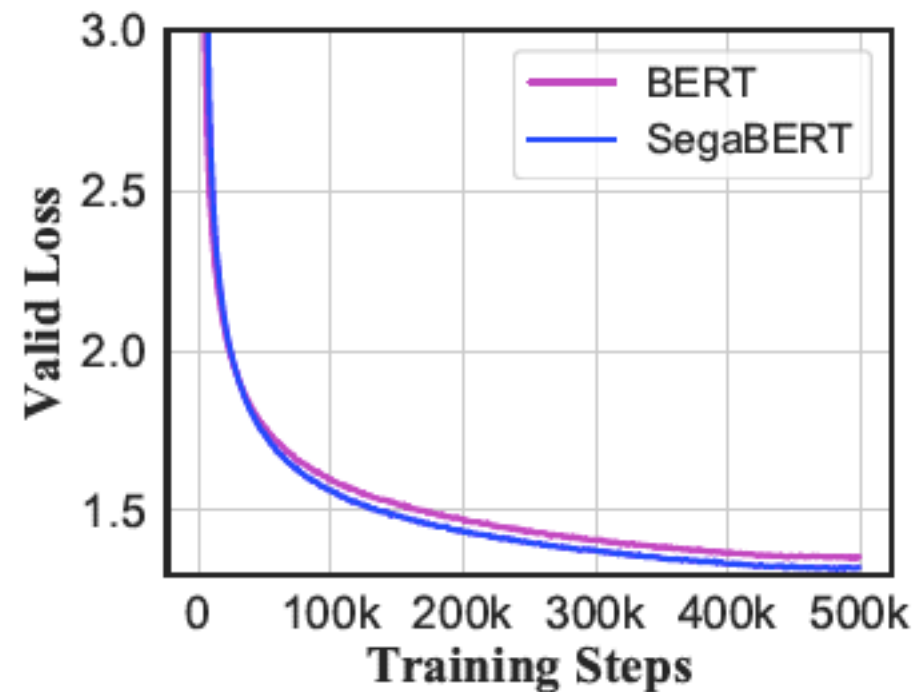


Figure 1: Input Representation of SegaBERT

Conversion speed much faster :



(a) Language modeling perplexities



(b) Pre-training losses

Figure 2: Valid perplexities and losses during the training processes of language modeling and pre-training.

Testing on GLUE dataset

Task(Metrics)	BASE model(wikipedia 500K steps)				LARGE model(wikibooks 1000K steps)			
	dev		test		dev		test	
	BERT	SegaBERT	BERT	SegaBERT	BERT	SegaBERT	BERT	SegaBERT
CoLA (Matthew Corr.)	55.0	54.7	43.5	50.7	60.6	65.3	60.5	62.6
SST-2 (Acc.)	91.3	92.1	91.2	91.5	93.2	94.7	94.9	94.8
MRPC (F1)	92.6	92.4	88.9	89.3	-	92.3	89.3	89.7
STS-B (Spearman Corr.)	88.9	89.0	83.9	84.6	-	90.3	86.5	88.6
QQP (F1)	86.5	87.0	70.8	71.4	-	89.1	72.1	72.5
MNLI-m (Acc.)	83.2	83.8	82.9	83.5	86.6	87.6	86.7	87.9
MNLI-mm (Acc.)	83.4	84.1	82.8	83.2	-	87.5	85.9	87.7
QNLI (Acc.)	90.4	91.5	90.1	90.8	92.3	93.6	92.7	94.0
RTE (Acc.)	68.3	71.8	65.4	68.1	70.4	78.3	70.1	71.6
Average	82.2	82.9	77.7	79.2	-	86.5	82.1	83.3

Table 2: The results on GLUE benchmark. All base models are pre-trained by this work. Every result of the dev set is the average score of 4 times finetuning with different random seeds. Scores of BERT large dev are from (Sun et al., 2019) and scores of BERT large test are from (Devlin et al., 2018).

Reading comprehension – SQUAD tasks

System	Dev	
	EM	F1
BERT base (Single)	80.8	88.5
BERT large (Single)	84.1	90.9
BERT large (Single+DA)	84.2	91.1
KT-NET	85.2	91.7
StructBERT Large (Single)	85.2	92.0
SegaBERT base (Single)	83.2	90.2
SegaBERT large (Single)	85.3	92.4

Table 3: Evaluation results of SQUAD v1.1.

System	Dev	
	EM	F1
BERT base	72.3	75.6
BERT base (ours)	75.4	78.2
SegaBERT base	76.3	79.2
BERT large	78.7	81.9
BERT large wwm	80.6	83.4
SegaBERT large	81.8	85.2

Table 4: Evaluation results of SQUAD v2.0.

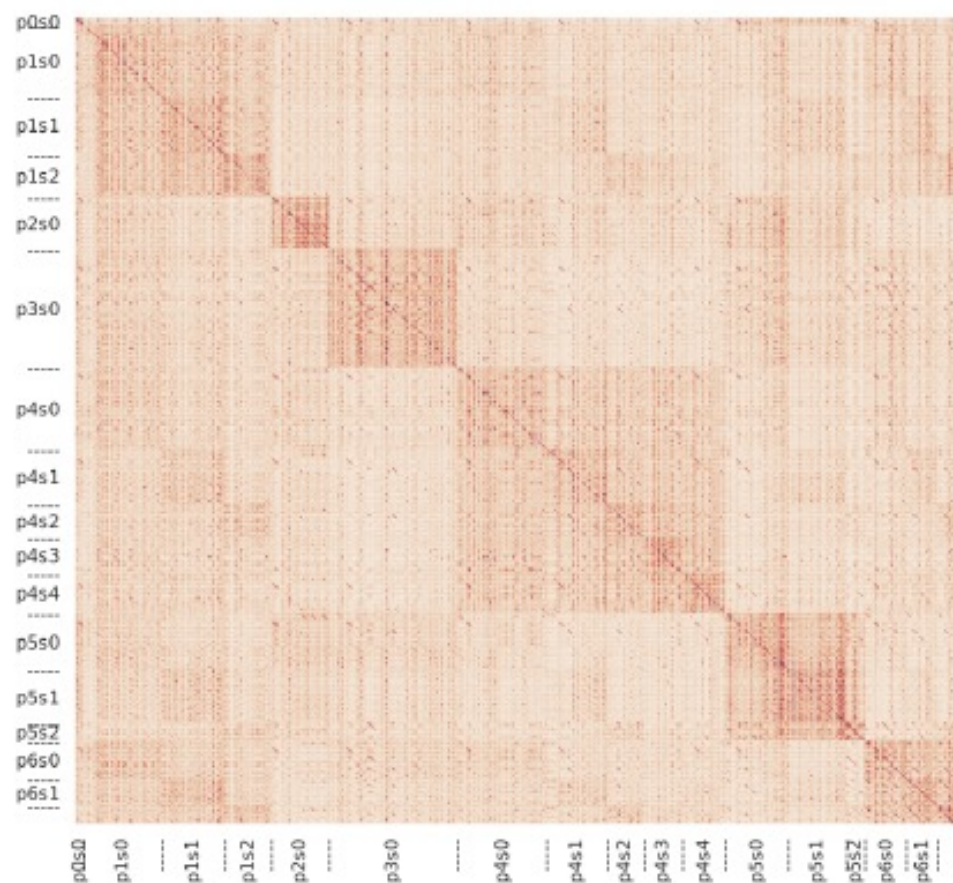
$F1 = 2 (P \cdot R) / (P + R)$, P is precision, R is recall, all in percentage, EM – exact match

Improving Transformer-XL

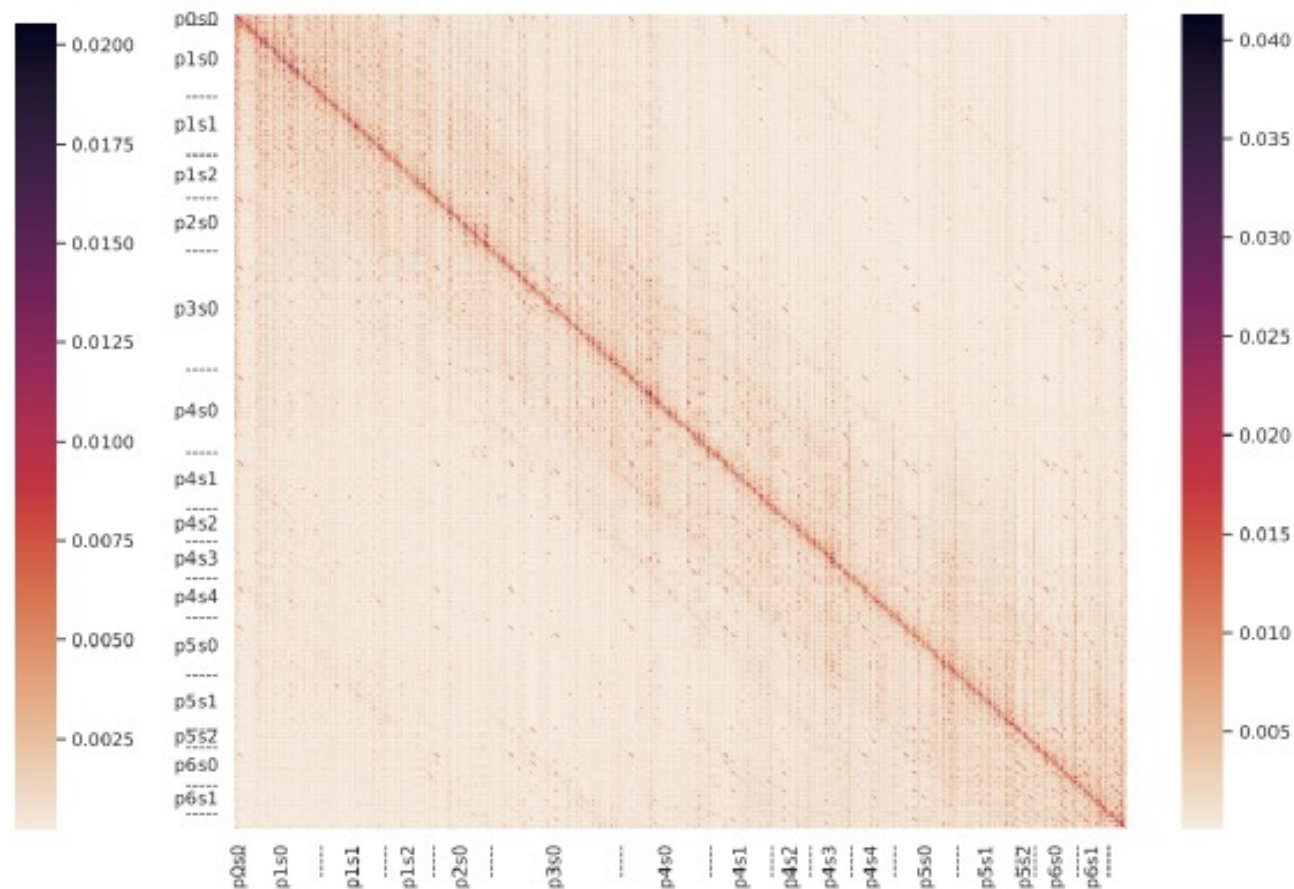
Model	#Param.	PPL
LSTM+Neural cache (Grave et al., 2017)	-	40.8
Hebbian+Cache (Rae et al., 2018)	-	29.9
Transformer-XL base, M=150 (Dai et al., 2019)	151M	24.0
Transformer-XL base, M=150 (ours)	151M	24.4
SegaTransformer-XL base, M=150	151M	22.5
Adaptive Input (Baevski and Auli, 2019)	247M	18.7
Transformer-XL large, M=384 (Dai et al., 2019)	257M	18.3
Compressive Transformer, M=1024 (Rae et al., 2020)	257M	17.1
SegaTransformer-XL large, M=384	257M	17.1

Table 1: Comparison with Transformer-XL and competitive baseline results on WikiText-103.

Looking at Attention

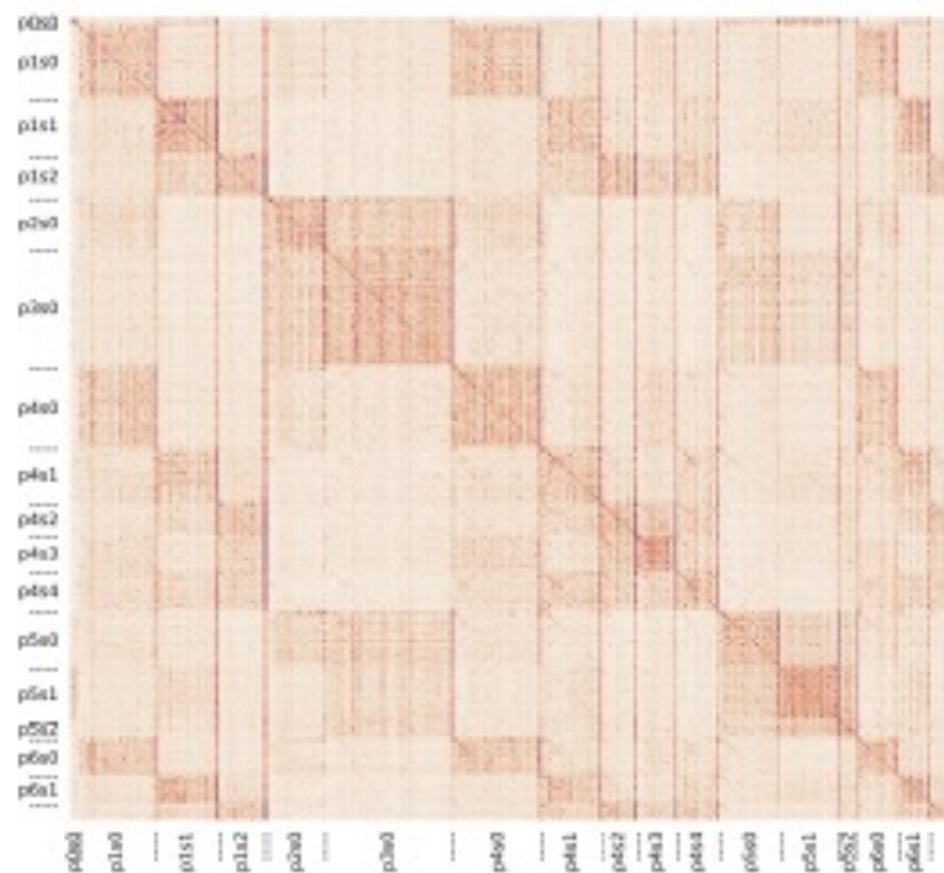


(a) SegBERT-Layer 1

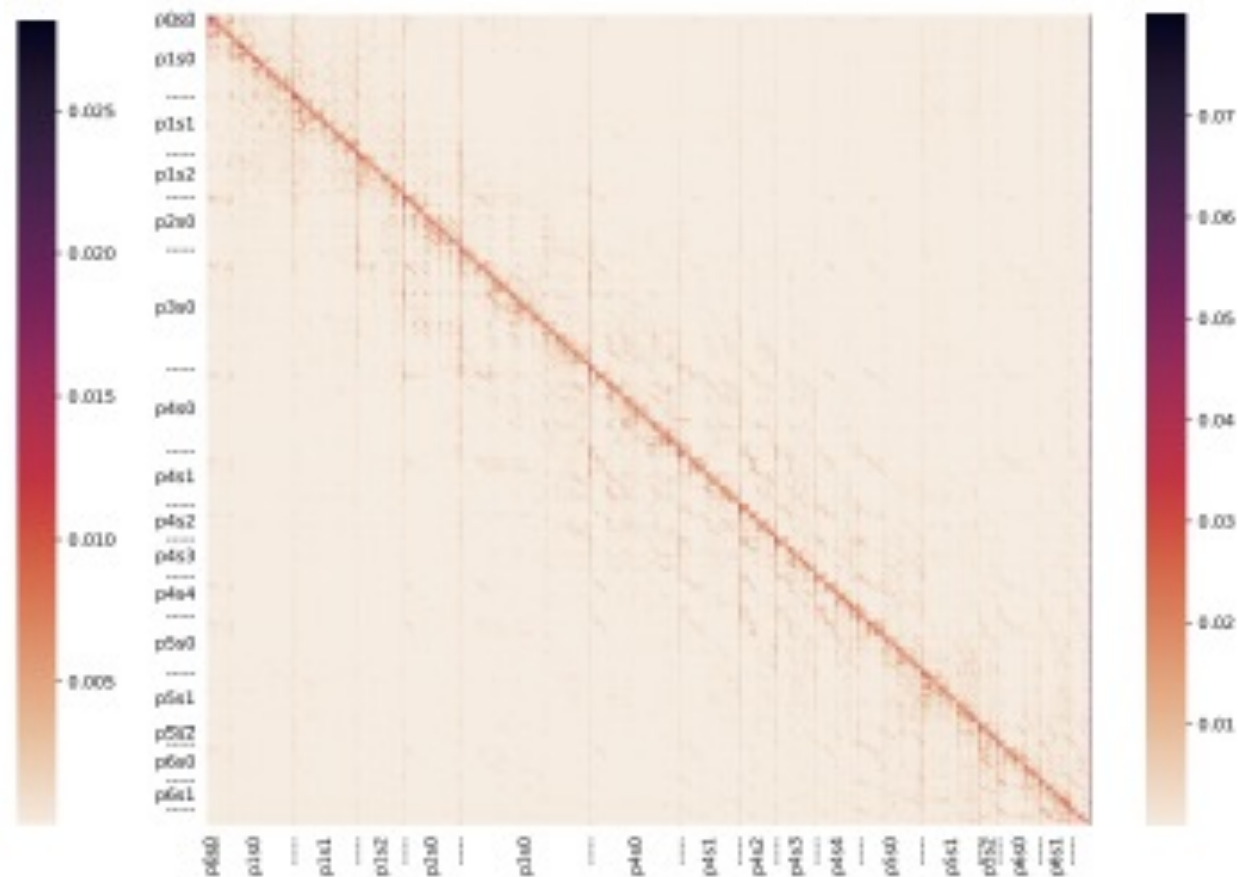


(b) BERT-Layer 1

Looking at Attention

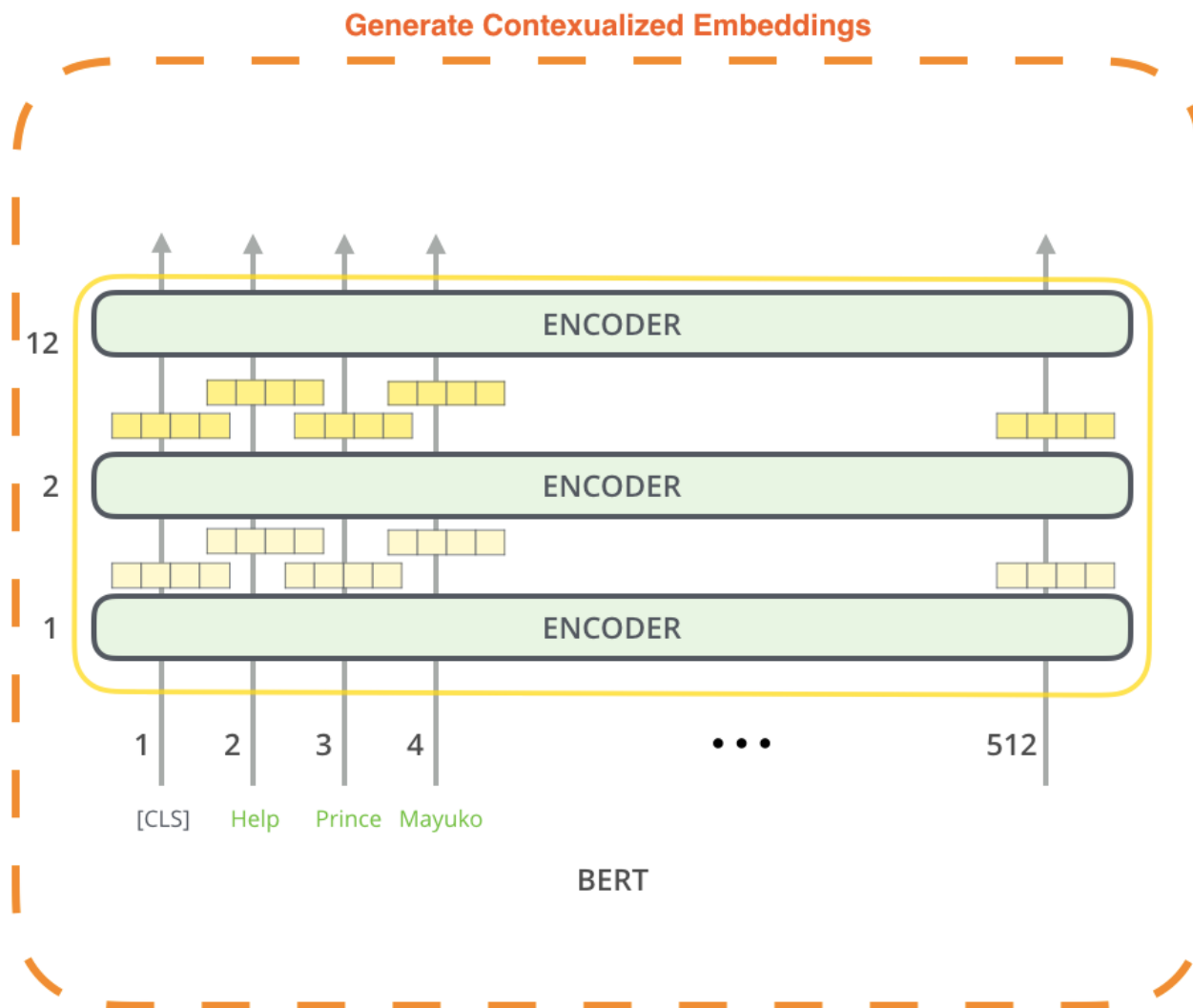


(a) SegBERT-Layer 6

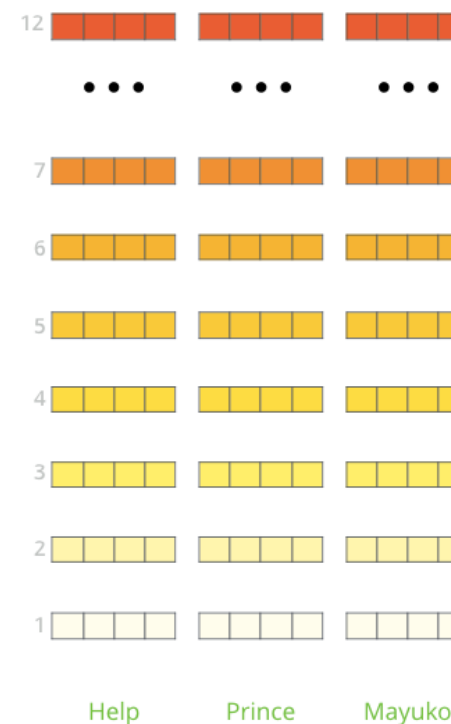


(b) BERT-Layer 6

Feature Extraction



The output of each encoder layer along each token's path can be used as a feature representing that token.



We end up with some embedding for each word related to current input

We start with independent word embedding at first level

But which one should we use?

What is the best contextualized embedding for “Help” in that context?

For named-entity recognition task CoNLL-2003 NER

		Dev F1 Score
12	First Layer	91.0
...	Last Hidden Layer	94.9
7	Sum All 12 Layers	95.5
6		
5		
4		
3	Second-to-Last Hidden Layer	95.6
2	Sum Last Four Hidden	95.9
1		
	Concat Last Four Hidden	96.1

Help



What we have learned

1. Model size matters (345 million parameters is better than 110 million parameters).
2. With enough training data, more training steps imply higher accuracy
3. Key innovation: Learning from unannotated data.
4. In biotechnology, we also have a lot of such data (for example meta-genomes).



Literature & Resources for Transformers

Resources:

OpenAI GPT-2 implementation: <https://github.com/openai/gpt-2>

BERT paper: J. Devlin et al, BERT, pretraining of deep bidirectional transformers for language understanding. Oct. 2018.

ELMo paper: M. Peters, et al, Deep contextualized word representation, 2018

ULM-FiT paper: Universal language model fine-tuning for text classification. J. Howear, S. Ruder., 2018

Jay Alammar, The illustrated GPT-2, <https://jalammar.github.io/illustrated-gpt2/>